Authors: Mayan Adler & Shlomi Hazan

Literature Review: Voice Spoofing Detection

Advanced Topics in Machine Learning • Stage 1

1. Abstract

The evolution of modern Text-to-Speech (TTS) and Voice Conversion (VC) systems has led to substantial improvements in synthetic speech. The emergence of neural codec-based approaches, exemplified by EnCodec and VALL-E, has substantially advanced the fidelity and naturalness of synthetic speech, bringing its acoustic characteristics increasingly close to those of authentic human speech. This review synthesizes twelve key papers analyzing the challenges of Audio Anti-Spoofing, emphasizing the generalization gap associated with previously unseen, out-of-distribution (OOD) generation methods. By examining novel datasets, advanced feature extraction, classification architectures, and Anomaly Detection paradigms, the review establishes a theoretical and empirical foundation for developing a robust, generalization-aware audio anti-spoofing classifier.

2. The Generalization Gap and Empirical Risk

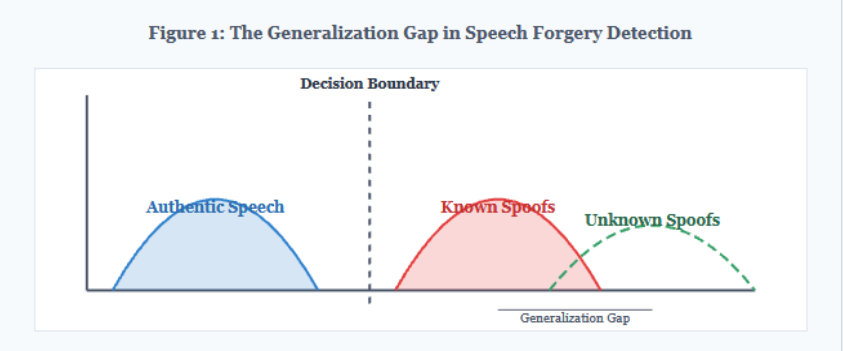
The core challenge in audio anti-spoofing is developing models that can generalize effectively to out-of-distribution (OOD) attacks. Statistical learning theory formalizes this: standard training minimizes the Empirical Risk over known anomalies, defined as:

$$R_{emp}(h) = (\frac{1}{n}) \sum_{i=1}^{n} L(h(xi), yi)$$

However, the continuous evolution of attack techniques means that the training distribution may no longer accurately represent the **true risk**, defined as the expected loss over the true data distribution:

$$R(h) = E_{(x,y)\sim P}[L(h(x), y)]$$

This discrepancy, commonly referred to as the "Generalization Gap", can cause models overfitted to conventional vocoder artifacts to exhibit substantial performance degradation when exposed to modern Neural Codecs or unfamiliar acoustic environments.



Visual representation of how minimizing empirical risk on known spoofing distributions (red) fails to capture novel, unseen attack distributions (green), highlighting the generalization gap.

3. Evaluation Frameworks and Dataset Diversity

Mitigating this gap requires datasets that capture diverse OOD acoustic conditions. The ASVspoof 5 challenge [1] introduces adversarial attacks (e.g., Malafide) and the architecture-agnostic Detection Cost Function (a-DCF) to account for high-confidence detection errors on unseen attack methods. The generalization challenge is evident in the performance of the state-of-the-art AASIST model, which yielded an EER of 29.12%, compared to 2.59% for adapted open systems. Similarly, CodecFake addresses neural codec-based synthesis, demonstrating EERs as low as 0.4% against specific VALL-E attacks. The AT-ADD benchmark further examines detector robustness under realistic distribution shifts, including compression, resampling, and replay effects. Xie et al. [2][12] likewise demonstrated substantial performance degradation under evaluation shifts, with FT-XLSR-AASIST declining from 99.70% to 76.73% Macro-F1, highlighting how aggregate validation metrics may conceal vulnerabilities to unseen generators such as BigVGAN.

Beyond attack diversity, source diversity is equally important. The Multi-Language Audio Anti-Spoofing Dataset (MLAAD) [10] addresses linguistic bias through more than 1,150 hours of synthetic speech across 54 languages, using M-AILABS as its source corpus. Cross-dataset evaluations highlight the complementary benefits of heterogeneous training sources, emphasizing the importance of linguistic and architectural diversity for OOD robustness. Furthermore, PhonemeDF [11] addresses the limitations of whole-recording classification through an MFA-aligned dataset of parallel genuine and synthetic recordings. Symmetric Kullback-Leibler Divergence (KLD) analysis reveals that synthesis artifacts tend to concentrate in specific phonetic units, such as diphthongs and fricatives. However, correlations between KLD and classification accuracy vary across feature representations, being generally positive for spectral features such as LFCC but inconsistent in Self-Supervised Learning (SSL) representations such as WavLM. These findings indicate that localized distributional divergence does not necessarily translate into improved classification performance.

4. Feature Representation and Classification Architectures

The effectiveness of audio anti-spoofing systems depends on the interaction between acoustic representations and classification backends. While conventional STFT features may fail to capture certain high-frequency artifacts, Constant Q Cepstral Coefficients (CQCC) [3] provide variable time-frequency resolution, enabling the detection of non-linear synthesis distortions. In deep learning, Self-Supervised Learning (SSL) models have demonstrated strong potential for extracting rich acoustic representations from large-scale unlabeled audio. wav2vec 2.0 [4] employs contrastive learning over quantized latent representations,

while WavLM [5] extends this framework through 94,000 hours of noisy and overlapped speech pre-training, incorporating a denoising objective that enhances representation robustness and supports speech anomaly detection.

Architecturally, graph-based models such as AASIST [6] capture long-range spectro-temporal dependencies via Heterogeneous Stacking Graph Attention Layers (HS-GAL), achieving an EER of 0.83%. Alternatively, hybrid architectures by Izrailov et al. [7] integrate classical features (MFCC, PLP) with Gaussian Mixture Model-Universal Background Models (GMM-UBM) and compact CNNs, reducing computational complexity while achieving an EER of 4.79%.

5. Generalization Paradigms and Anomaly Detection

To reduce reliance on labeled spoofing samples, Yin and Lei [8] combine wav2vec 2.0 and LFCC features within a Deep Support Vector Data Description (Deep SVDD) architecture. The model learns a compact hypersphere around genuine speech representations by minimizing their distance from a predefined center c:

$$min_W(\frac{1}{n}) \sum_{i=1}^{n} ||f(xi;\ W)\ -\ c||^2\ +\ (\frac{\lambda}{2})||W||^2$$

Consequently, OOD recordings falling outside this boundary can be identified as potential anomalies, achieving an EER of 6.89% against unknown attacks. In contrast, Wu et al. [9] employ a Vector-Quantized Variational Autoencoder (VQ-VAE) to enhance discriminative spoofing features. Guided by True Speech Consistency and Triplet Loss objectives, the framework promotes greater separation between genuine and synthetic speech representations, reducing EER from 22.64% to 14.04% using conventional spectral features.

6. Conclusion and Baseline Transition

Addressing voice spoofing vulnerabilities requires the integration of diverse datasets, advanced acoustic representations (e.g., WavLM), and anomaly detection approaches such as Deep SVDD. In Stage 2, we will establish a baseline using conventional MFCC features and Logistic Regression to empirically quantify performance degradation under previously unseen, out-of-distribution (OOD) attacks. Following the methodological principles of AT-ADD, particularly controlled evaluation under distribution shifts, this baseline will provide a rigorous quantitative benchmark. Building on these results, we will investigate the transition to SSL-based architectures and incorporate complementary subsets from MLAAD and PhonemeDF into the training pipeline to improve generalization and better approximate the true risk under real-world conditions.

References

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